

Qns	Answer	Marks
7(a)	$t = 1.2 \text{ s}$ $E = \frac{d(N\phi)}{dt} = NA \frac{dB}{dt} = \frac{560 \times \pi \times 0.045^2 \times 0.250}{1.2}$ $= 0.74 \text{ V or } 0.742 \text{ V}$ (accept 2 s.f. or 3 s.f.)	C1 A1
7(b)	Sheet <u>cuts</u> magnetic flux / experiences a <u>changing magnetic flux</u> (linkage) and causes <u>induced e.m.f.</u> in sheet (induced) e.m.f. causes (<u>induced eddy</u>) <u>currents</u> (in sheet) Option 1: Currents (in sheet is in a magnetic field which) <u>cause resistive force</u> Option 2: Currents (in sheet) <u>dissipate (thermal) energy</u>. (Thermal energy comes from energy of oscillations) Option 3: Currents (in sheet) produce a magnetic field around sheet which is in opposite direction to the external magnetic field and the <u>sheet is repelled</u> (current cannot pass all the way around the sheet / path of current is broken / not a complete/closed loop) Option 1: Smaller currents in Y OR smaller resistive force in Y Option 2: Larger currents in X OR larger resistive force in X, so dashed line is X	B1 B1 B1 B1

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Q(2)(i)	Label V/ between 400-700 nm on the horizontal axis	D1